

Client Brief:

This report has been commissioned to answer the following:

- Assess customer attitudes towards COVID19 and implications for the retail sector: What is important to customers?
- What recommendations do you have for Countdown as a large retailer in New Zealand to act on these attitudes and beliefs?

Executive Summary

The global pandemic of the COVID-19 virus has heavily impacted the global economy. The purpose of this report is to convert the COVID-19 survey data into meaningful information that recognises any patterns or opportunities relevant to the retail sector. The information will be specifically looking at customer attitudes towards COVID-19, what is important to customers and the implications this has for the retail sector. Guidance in how to take direction towards informed and effective business decisions from interpreting the data will be provided. Recommendations for action based on the interpretations and current customer attitudes towards COVID-19 will be made.

Key Findings

- Customers feel safer in the presence of protective measures against COVID19 such as face coverings
- The influence of COVID19 on customer perceptions and behaviour is situational
- COVID19 has increased demand for online shopping
- COVID19 decreases the frequency with which consumers shop and the length of time they shop for

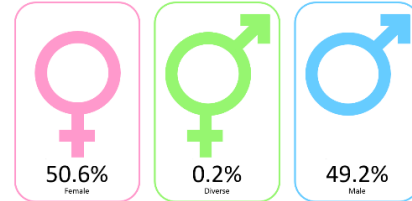


Demographic profile

Demographic information of respondents was collected to determine or rule out whether these factors influenced the behaviours.

Gender

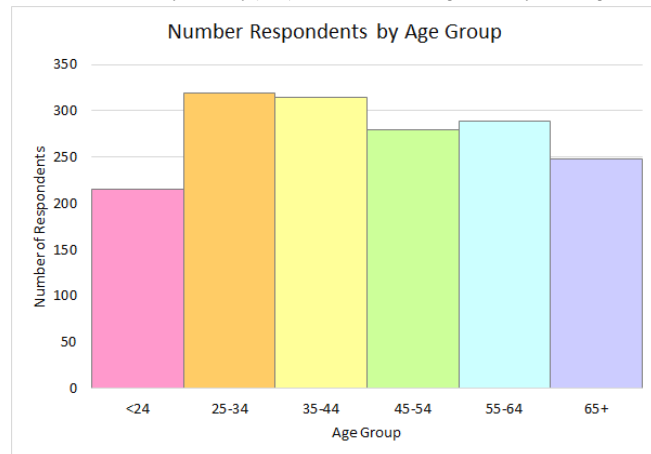
Of all respondents 50.6% identify as female, 49.2% identify as male and the remaining 0.2% identify as gender diverse.



Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Age

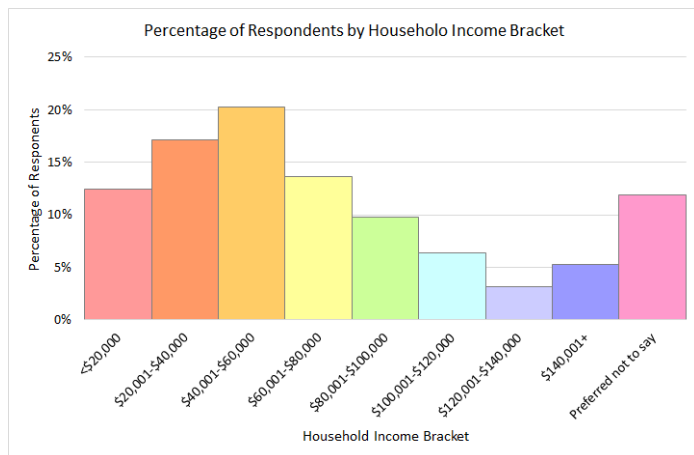
The age group with the most respondents is the 25-34 range with 319 respondents. The less than 24 age group had the fewest respondents with only 215. Of the 1667 respondents to this survey, 1 respondent did not enter their age.



Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Income

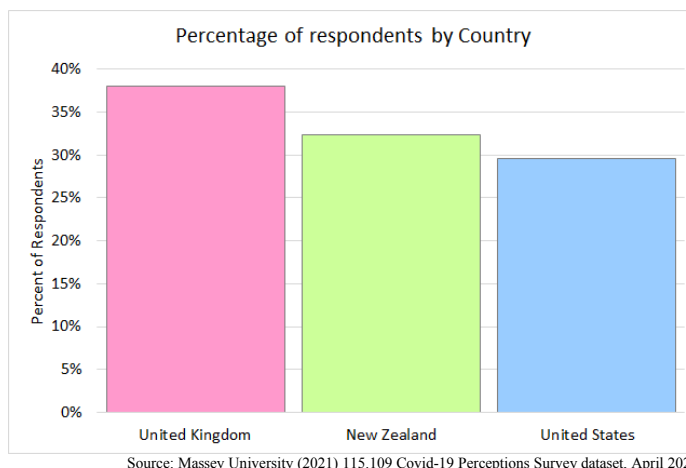
The income bracket with the largest percentage of respondents is the \$40,001-\$60,000 bracket with 20% respondents selecting this range. The \$120,001-\$140,000 income bracket smallest portion of respondents at 3%. Given the option, 12% of the survey's respondents preferred not to say which household income bracket they fit.



Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Country of Respondent

The survey has respondents from three countries. At 38% the largest portion of respondents are from the United Kingdom. Then New Zealand with 32% of respondents, the remaining 30% of respondents come from the United States.

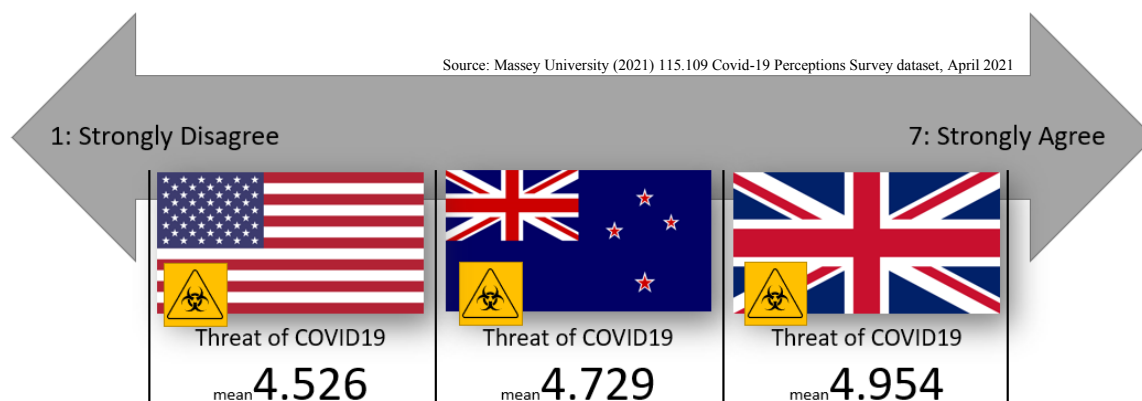


Perceptions of COVID19

The survey investigated respondents' perceptions of COVID19 as a threat in general, degree of support for face mask use, and their agreement with various reasons for wearing masks.

Threat Level

Respondents were asked to rate their level of agreement with a series of statements related to how dangerous they feel COVID19 is in general. The responses to these statements were collated together to depict how threatening COVID19 is perceived to be without a situational guide. Agreement level was ranked from 1: strongly disagree to 7: strongly agree, a higher level of agreement equates to a higher perceived threat level.

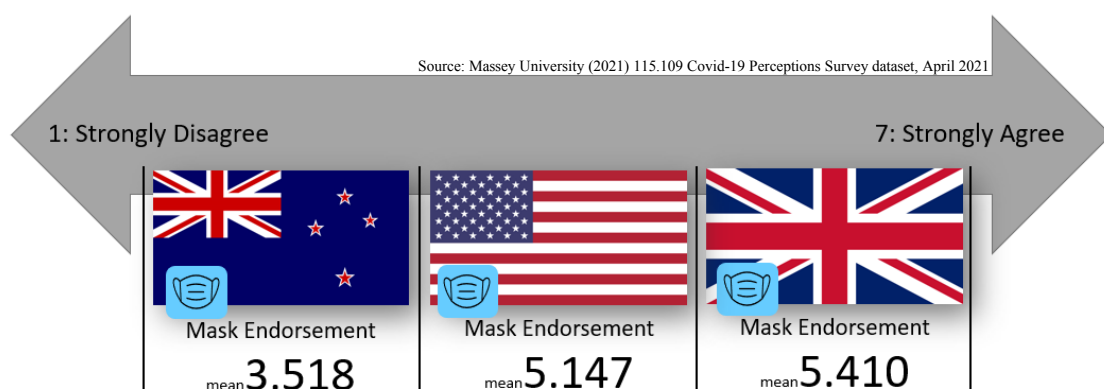


The figure above depicts the mean of responses from each country ranked from lowest to highest. The null hypothesis that general threat level and a respondent's country were independent was rejected by all t-Tests conducted on the differences between the means of respondents from each of the different countries (Appendix A).

Mask Endorsement

Respondents were asked to rate their level of agreement with a series of statements related to how important they feel face masks are within a variety of situations. The responses to these statements were collated together to depict a respondent's overall face mask endorsement. Agreement level was ranked from 1: strongly disagree to 7: strongly agree, a higher level of agreement equates to a higher degree of endorsement for face masks.

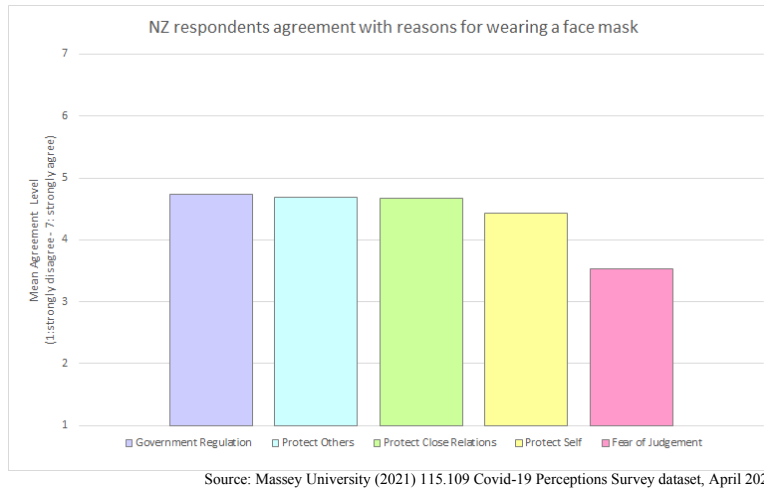
The figure below depicts the mean of responses from each country ranked from lowest to highest. The null hypothesis that the variables mask endorsement and a respondent's country are independent was rejected by t-Tests conducted on the differences between each of these means (Appendix A).



Reasons for wearing face Masks

Respondents were asked to rate their level of agreement with a series of statements related to reasons for wearing face masks. Agreement level was ranked from 1: strongly disagree to 7: strongly agree, a higher level of agreement equates to a higher degree of endorsement for face masks.

The figure below depicts the mean of responses from New Zealand respondents ranked from highest to lowest. The null hypothesis, that the difference in means between government regulation, protecting others, and protecting close relations is not meaningful was not rejected. However, it was rejected for the difference in means between protecting close relations and protecting self; and between protecting self and fear of judgement (Appendix B).



Thus, the final order of reasons for New Zealand respondents is

1. Government Regulation, Protect Others, Protect Close Relations
2. Protect Self
3. Fear of Judgement.

Environmental Influences

Scenario

The survey separately assessed situational threat level using a contextual guide and two control groups. The scenario aimed to investigate and align whether the strength of emotions respondents purported they would have aligned with what they responded when presented with a situational guide. The situational guide was a picture of a supermarket shopping isle showing two people, in the first control group the people were not wearing face masks, in the second they were wearing face masks. Respondents were asked how threatening they perceived the situation to be. The responses were then controlled depending on whether the scenario depicted shoppers wearing face coverings or not.

Situational Threat Level

Agreement was again ranked from 1-7 with a higher level of agreement equating to a higher perceived threat level. The mean response to these questions from respondents in the control group not shown a mask was higher than the mean response of respondents in the second control group that was shown a mask. The difference in these means has a t-test score of 8.075, given the critical value of 1.961 the null hypothesis that these variables are independent is rejected (Appendix C). The alternative hypothesis that they are not independent is accepted meaning the difference in their means is meaningful.



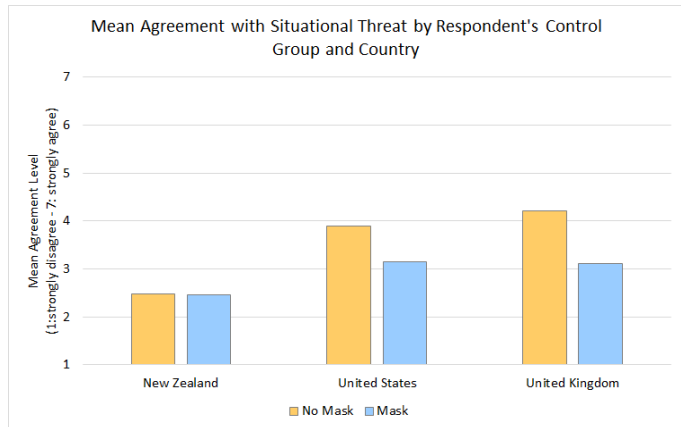
Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

However, this difference in mean response is not seen to the same extent in New Zealand respondents. A t-Test of the difference in mean confirms that the difference in mean situational threat between New Zealand respondents not shown the mask and New Zealand respondents shown the mask is not statistically meaningful (Appendix D).

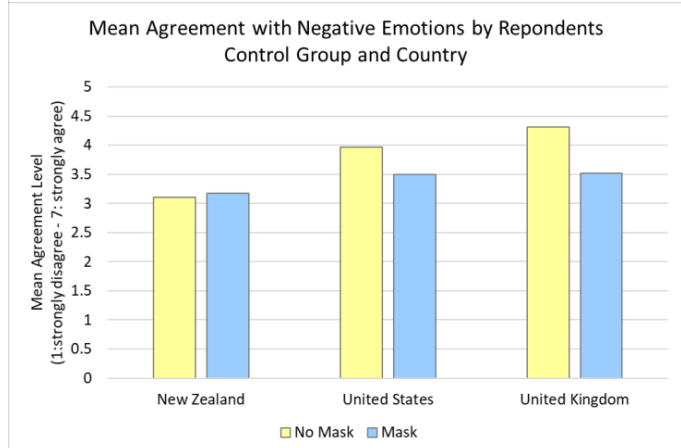
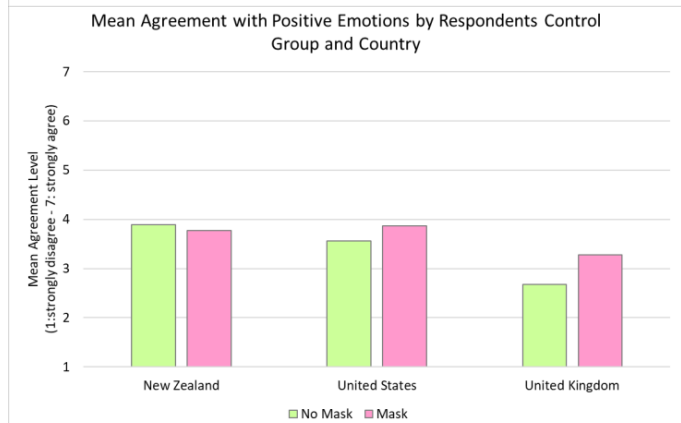
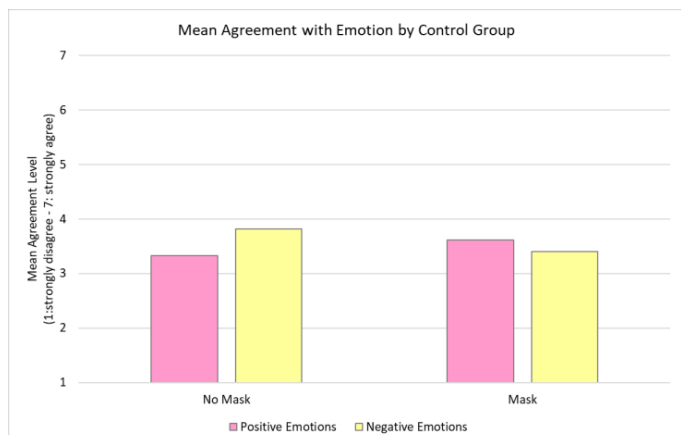
Emotions

Respondents in the first control group, who were not shown a mask, displayed a higher mean agreement with negative emotions and a lower mean agreement with positive emotions than respondents from the second control group who were shown a mask. The t-Tests conducted on these differences in mean both rejected the null hypothesis. The alternate hypothesis, that the difference in these means is meaningful is accepted instead (Appendix E).

These results were again not seen to the same extent in New Zealand respondents. A t-Test of the difference in mean confirms that the difference in mean agreement with both positive and negative emotions between New Zealand respondents not shown the mask and New Zealand respondents shown the mask is not statistically meaningful (Appendix F).



Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021



Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Shopping Behaviours

The mean response to each of the composite shopping behaviour variables differed between the two control groups. These differences were determined to be statistically meaningful for all variables except the buy more items and buy less items composites (Appendix G).



Conclusions

The statistically meaningful difference in mean situational threat perceived by respondents from the different control groups suggests customers feel safer in the presence of protective measures against COVID19 such as face coverings. This is further supported by the meaningful differences between the mean emotional response to the presented scenario of each control group. Respondents presented the image with protective face coverings were more likely to agree with positive emotions and less likely to agree with negative emotions. While the reverse is true for respondents presented an image that did not depict protective face coverings. Untaru and Kan (2021) demonstrate that the presence of protective measures positively influences customer satisfaction which in turn influences customer intentions.

When situational threat and emotional response were further separated by respondent country it was found that the difference between control groups of New Zealand respondents was not statistically significant while the difference between control groups of United States and United Kingdom respondents was. Doogan et al. (2020) highlighted that public perception of COVID19 in New Zealand may be influenced by the rapidity with which the virus was controlled. Given this, and that the control groups were implemented to assess the situational influence of COVID19 on consumer perceptions and behaviour, the meaningful difference in means between control groups for respondents from other countries will be considered when making recommendations.

A statistically meaningful difference between the means of control groups was found with their responses to various shopping behaviours. Respondents in the control group not shown a mask were more likely to agree with the online shopping propensity, less time in store, and lower shopping frequency composites. While respondents in the second control group that was shown a mask were less likely to agree with these behaviours and more likely to agree with more time in store. There was no statistically meaningful difference between the means of the control groups for the purchasing more items and purchasing less items composites. Vázquez-Martínez et al. (2021) illustrates the key role COVID19 has had in driving consumer behaviour towards online environments. Kim (2020) emphasises the opportunity it has presented in the online space by highlighting the increase in online sale demand.

Recommendations for Countdown in NZ

- Given the importance of situational context upon whether COVID19 will influence consumer perception of behaviour it is recommended that Countdown ensure that visibility of protection measures increase and decrease concurrently with the presence of COVID19.
- Creating service process plans that are tailored to address the differing consumer demands and shopping behaviours of consumers when COVID19 is prevalent vs when it isn't.
- Increases in consumer demand for online services present an opportunity for Countdown to refine and expand their current online experience.

References

- Doogan, C., Buntine, W., Linger, H., & Brunt, S. (2020). Public perceptions and attitudes toward COVID-19 nonpharmaceutical interventions across six countries: A topic modeling analysis of Twitter data. *Journal of Medical Internet Research*, 22(9). <https://doi.org/10.2196/21419>
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- Untaru, E.-N., & Han, H. (2021). Protective measures against COVID-19 and the business strategies of the retail enterprises: Differences in gender, age, education, and income among shoppers. *Journal of Retailing and Consumer Services*, 60. <https://doi.org/10.1016/j.jretconser.2021.102446>
- Vázquez-Martínez, U. J., Morales-Mediano, J., & Leal-Rodriguez, A. L. (2021). The impact of the COVID-19 crisis on consumer purchasing motivation and behavior. *European Research on Management and Business Economics*. <https://doi.org/10.1016/j.iedeen.2021.100166>

Appendices

Appendix A

t-Tests of differences between means of respondents from different countries for the General Threat of COVID19 and Mask Endorsement composite variables.

H_0 = The differences between the means of respondents from different countries are not statistically meaningful.

H_a = The differences between the means of respondents from different countries are statistically meaningful.

Variable Group	NZ Respondents			UK Respondents			t-Test (NZ v UK)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
General Threat COVID19	4.729	0.9571	539	4.954	0.9722	635	-3.964	1.9620	N
Mask Endorsement	3.518	1.5965	539	5.410	1.1530	635	-23.500	1.9620	N
Variable Group	NZ Respondents			US Respondents			t-Test (NZ v US)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
General Threat COVID19	4.729	0.9571	539	4.526	1.1442	493	3.107	1.9623	N
Mask Endorsement	3.518	1.5965	539	5.147	1.5597	493	-16.552	1.9623	N
Variable Group	UK Respondents			US Respondents			t-Test (US v UK)		
	Variable's Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
General Threat COVID19	4.954	0.9722	635	4.526	1.1442	493	6.779	1.9621	N
Mask Endorsement	5.410	1.1530	635	5.147	1.5597	493	3.255	1.9621	N

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix B

t-Tests of differences between means of reasons for wearing face masks for New Zealand

Respondents.

H_0 = The differences between means of tested reasons are not statistically meaningful for New Zealand Respondents.

H_a = The differences between means of tested reasons are statistically meaningful for New Zealand Respondents.

Respondent Group	Variable's Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-Test		
							t-test statistic	critical value of t	Accept Null Hypothesis?
NZ Respondents	Government Regulation			Protect Others			t-Test (Gov Reg v Others)		
	4.737	1.6856	539	4.685	1.8058	539	0.488	1.962	Y
	Protect Others			Protect Close Relationships			t-Test (Others v Close)		
	4.685	1.8058	539	4.677	1.8055	539	0.067	1.962	Y
	Protect Close Relationships			Protect Self			t-Test (Close v Self)		
	4.677	1.8055	539	4.430	1.8474	539	2.218	1.962	N
	Protect Self			Fear of Judgement			t-Test (Self v Judgement)		
	4.430	1.8474	539	3.532	1.7428	539	8.208	1.962	N
	Government Regulation			Protect Close Relationships			t-Test (Gov Reg v Close)		
	4.737	1.6856	539	4.677	1.8055	539	0.558	1.962	Y
NZ Respondents	Government Regulation			Protect Self			t-Test (Gov Reg v Self)		
	4.737	1.6856	539	4.430	1.8474	539	2.842	1.962	N
	Protect Others			Protect Self			t-Test (Others v Self)		
	4.685	1.8058	539	4.430	1.8474	539	2.284	1.962	N

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix C

t-Tests of differences between means of respondents from different control groups for the situational threat composite. The test is repeated for all respondents, and respondents of each country separately.

H_0 = The differences between means of the different control groups are not statistically meaningful.

H_a = The differences between means of the different control groups are statistically meaningful.

Variable Group	No Mask Shown			Mask Shown			t-Test (Mask v No Mask)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable's Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Situational Threat	3.563	1.8044	830	2.912	1.4720	837	8.075	1.961	N
Situational Threat NZ	2.477	1.3876	264	2.468	1.2495	275	0.080	1.964	Y
Situational Threat US	3.886	1.8279	249	3.145	1.6069	244	4.771	1.965	N
Situational Threat UK	4.214	1.6808	317	3.116	1.4597	318	8.785	1.964	N

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix D

t-Tests of differences between means of respondents from different countries for the Situational Threat composite variable. The test is repeated for each control group.

H_0 = The differences between the means of respondents from different countries are not statistically meaningful for the situational threat composite.

H_a = The differences between the means of respondents from different countries are statistically meaningful situational threat composite.

Variable Group	NZ Respondents			UK Respondents			t-Test (NZ v UK)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Situational Threat (No Mask)	2.477	1.3876	264	4.214	1.6808	317	-13.407	1.9641	N
Situational Threat (Mask)	2.468	1.2495	275	3.116	1.4597	318	-5.761	1.9640	N
Variable Group	NZ Respondents			US Respondents			t-Test (NZ v US)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Situational Threat (No Mask)	2.477	1.3876	264	3.886	1.8279	249	-9.863	1.9646	N
Situational Threat (Mask)	2.468	1.2495	275	3.145	1.6069	244	-5.391	1.9646	N
Variable Group	UK Respondents			US Respondents			t-Test (US v UK)		
	Variable's Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Situational Threat (No Mask)	4.214	1.6808	317	3.886	1.8279	249	2.218	1.9642	N
Situational Threat (Mask)	3.116	1.4597	318	3.145	1.6069	244	-0.224	1.9642	Y

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix E

t-Tests of differences between means of respondents from different control groups for the positive emotions and negative emotions composite variables. The tests are repeated for all respondents, and respondents of each country separately.

H_0 = The differences between means of the different control groups are not statistically meaningful.

H_a = The differences between means of the different control groups are statistically meaningful.

Variable Group	No Mask Shown			Mask Shown			t-Test (No Mask v Mask)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable's Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Positive Emotions	3.327	1.3703	830	3.613	1.1455	837	-4.628	1.961	N
Positive Emotions NZ	3.893	1.0282	264	3.767	1.0427	275	1.413	1.964	Y
Positive Emotions US	3.561	1.4179	249	3.872	1.1643	244	-2.661	1.965	N
Positive Emotions UK	2.671	1.3138	317	3.281	1.1395	318	-6.248	1.964	N
Negative Emotions	3.820	1.3514	830	3.398	1.1343	837	6.916	1.961	N
Negative Emotions NZ	3.103	1.1112	264	3.173	1.0456	275	-0.760	1.964	Y
Negative Emotions US	3.962	1.4483	249	3.497	1.2320	244	3.838	1.965	N
Negative Emotions UK	4.306	1.1976	317	3.515	1.1042	318	8.649	1.964	N

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix F

t-Tests of differences between means of respondents from different countries for the positive emotions and negative emotions composite variables. The test is repeated for each control group.

H_0 = The differences between the means of respondents from different countries are not statistically meaningful.

H_a = The differences between the means of respondents from different countries are statistically meaningful.

Variable Group	NZ Respondents			UK Respondents			t-Test (NZ v UK)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Positive Emotions (No Mask)	3.893	1.0282	264	2.671	1.3138	317	12.296	1.9641	N
Positive Emotions (Mask)	3.767	1.0427	275	3.281	1.1395	318	5.386	1.9640	N
Negative Emotions (No Mask)	3.103	1.1112	264	4.306	1.1976	317	-12.459	1.9641	N
Negative Emotions (Mask)	3.173	1.0456	275	3.515	1.1042	318	-3.855	1.9640	N
Variable Group	NZ Respondents			US Respondents			t-Test (NZ v US)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Positive Emotions (No Mask)	3.893	1.0282	264	3.561	1.4179	249	3.046	1.9646	N
Positive Emotions (Mask)	3.767	1.0427	275	3.872	1.1643	244	-1.090	1.9646	Y
Negative Emotions (No Mask)	3.103	1.1112	264	3.962	1.4483	249	-7.565	1.9646	N
Negative Emotions (Mask)	3.173	1.0456	275	3.497	1.2320	244	-3.235	1.9646	N
Variable Group	UK Respondents			US Respondents			t-Test (US v UK)		
	Variable's Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Positive Emotions (No Mask)	2.671	1.3138	317	3.561	1.4179	249	-7.726	1.9642	N
Positive Emotions (Mask)	3.281	1.1395	318	3.872	1.1643	244	-6.043	1.9642	N
Negative Emotions (No Mask)	4.306	1.1976	317	3.962	1.4483	249	3.091	1.9642	N
Negative Emotions (Mask)	3.515	1.1042	318	3.497	1.2320	244	0.188	1.9642	Y

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021

Appendix G

t-Tests of differences between means of respondents from different control groups for each of the shopping behaviour composites.

H_0 = The differences between means of the different control groups are not statistically meaningful.

H_a = The differences between means of the different control groups are statistically meaningful.

Variable Group	No Mask Shown			Mask Shown			t-Test (No Mask v Mask)		
	Variable Mean	Variable Standard Deviation	Variable Sample Size	Variable Mean	Variable Standard Deviation	Variable Sample Size	t-test statistic	critical value of t	Accept Null Hypothesis?
Online shopping propensity	3.996	1.6278	830	3.769	1.6346	837	2.841	1.961	N
More Time Shopping	3.514	1.3457	830	3.640	1.2428	837	-1.996	1.961	N
Less Time Shopping	4.479	1.6012	830	4.265	1.5535	837	2.765	1.961	N
Lower Shopping Frequency	4.358	1.5094	830	4.213	1.4409	837	2.017	1.961	N
More Items	3.599	1.6384	830	3.704	1.5345	837	-1.349	1.961	Y
Less Items	4.653	1.5626	830	4.550	1.5152	837	1.372	1.961	Y

Source: Massey University (2021) 115.109 Covid-19 Perceptions Survey dataset, April 2021